



Geometry

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · FLORIDA COURSE 1206310 · HONORS 1206320 · FULL YEAR (TWO SEMESTERS)

~4¼ hrs per week
50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in the lessons, examples, and practice sets from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Geometry trains scholars to reason carefully about space and shape. The course begins with the building blocks of geometry and a real study of logic — truth tables, conditional statements, and how a proof works — then develops the classical results on triangles, congruence, similarity, trigonometry, and quadrilaterals. The second semester turns to circles, measurement of two- and three-dimensional figures, coordinate geometry, and transformations. Proof and precise argument run through the whole year.

THE YEAR AT A GLANCE

M1 Introduction to Geometry (~2 wks)
points, lines, angles, constructions

M3 Triangle Relationships & Congruency (~2½ wks)
congruence criteria and proofs

M5 Trigonometry (~2 wks)
sine, cosine, tangent, special right triangles

M7 Circles (~2 wks)
chords, tangents, arcs, inscribed figures

M9 Coordinate Geometry (~2½ wks)
distance, midpoints, classifying figures

M11 Advanced Geometric Applications (~1½ wks)
cross sections, equations of circles

M2 Logical Reasoning (~2½ wks)
truth tables, conditionals, first proofs

M4 Similarity & Proportional Relationships (~1½ wks)
similar triangles, geometric mean

M6 Quadrilaterals (~2½ wks)
parallelograms, trapezoids, classification proofs

M8 Geometric Measurements (~2 wks)
area, surface area, volume, scale factors

M10 Transformations (~3 wks)
reflections, rotations, translations, dilations

Honors (1206320). Honors scholars add proof by contradiction, the Laws of Sines and Cosines, additional compass-and-straightedge constructions, and formal justification of the congruence and similarity criteria.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- A review and module assessment as each module closes, roughly every two to three weeks.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific calculator (for the trigonometry work) plus a compass, straightedge, and protractor for constructions.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.